

MH-Neurons: Do Modality-Specialized Neurons Drive Hallucination in VLMs? A Neuron Taxonomy Perspective

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Abstract. Vision-language models (VLMs) process visual and textual information through shared transformer layers, yet whether modality-specialized neurons disproportionately drive hallucination remains poorly understood. We term such neurons MH-Neurons (Modality-specialized Hallucination Neurons). Prior work [15] classifies neurons using fixed activation thresholds with four hand-tuned hyperparameters, producing a large multimodal category that conflates genuinely cross-modal neurons with ambiguous cases. We propose the Permutation-Based Modality Test (PMBT), a non-parametric alternative replacing all four thresholds with a single significance level. Across four architectures, PMBT reclassifies 15–32% of neurons labelled multimodal by our reimplementations of Xu *et al.*’s [15] fixed threshold method. Ablation confirms a double dissociation between visual and text neuron function. Enrichment analysis on LLaVA-OneVision-7B, InternVL2.5-8B, and Qwen2.5-VL-7B shows visual neurons are over-represented among hallucination drivers (fold 1.10–1.35 $\times$, $p < 0.001$). Steering on LLaVA-OneVision-7B at $\alpha = 0.5$ improves POPE from 90.2% to 92.0% and reduces CHAIR_i from 26.1% to 24.3%.

Keywords: Vision-language models · Neuron interpretability · Modality classification · Hallucination · Activation steering

1 Introduction

Vision-language models (VLMs) such as LLaVA [11], InternVL [3], and Qwen-VL [1] have achieved remarkable progress on tasks spanning visual question answering, image captioning, and multimodal reasoning. These models share a common architecture: a vision encoder projects image patches into visual tokens that are concatenated with text tokens and processed by an autoregressive LLM backbone. Despite their success, VLMs frequently hallucinate, generating plausible but factually incorrect descriptions of visual content, raising concerns about their reliability in high-stakes applications.

Understanding which neurons in the LLM backbone are responsible for processing visual versus textual information is a prerequisite for diagnosing and mitigating such failures. Xu *et al.* [15] pioneered this direction by classifying neurons as visual, text, or multimodal based on their activation patterns across

modalities. However, their method relies on four hand-tuned thresholds (activation magnitudes T_v , T_t and minimum token counts n_v , n_t) that lack statistical justification. This produces a large multimodal category (15–45% of all neurons) that conflates genuinely cross-modal neurons with ambiguous cases where both modalities weakly activate.

In this paper, we address three questions: (1) Can we replace the ad-hoc thresholds with a statistically principled classification? (2) Does the resulting taxonomy capture functionally meaningful distinctions, as verified by causal intervention? (3) Are specific modality types disproportionately responsible for hallucination, and can targeted suppression reduce it? We introduce the term **MH-Neurons** (Modality-specialized **H**allucination **N**eurons) to denote neurons that are both confidently classified as modality-specific by our permutation test and identified as hallucination-driving by causal ablation scoring.

To answer these questions, we propose a pipeline spanning neuron classification, causal validation, hallucination neuron identification, and activation steering (Fig. 1). At its core is the Permutation-Based Modality Test (PMBT), which replaces all four fixed thresholds with an adaptive per-neuron threshold (via Otsu’s method) and a non-parametric permutation test (1,000 permutations, $\alpha = 0.05$). PMBT provides a calibrated p -value per neuron, eliminating the need for hand-tuning.

We evaluate PMBT on four VLM architectures spanning different backbones and vision encoders: LLaVA-1.5-7b (LLaMA-2, CLIP), LLaVA-OneVision-7B (Qwen2, SigLIP), InternVL2.5-8B (InternLM2, InternViT), and Qwen2.5-VL-7B (Qwen2, native ViT). LLaVA-1.5-7b additionally serves as a validation model for direct comparison with Xu *et al.* [15]. Across all four models, PMBT reclassifies 15–32% of neurons labelled multimodal by our reimplement of Xu *et al.*’s [15] fixed threshold method. Ablation experiments establish a double dissociation: visual neuron ablation degrades POPE and CHAIR while preserving TriviaQA and MMLU, and vice versa for text neurons. Enrichment analysis on LLaVA-OneVision-7B, InternVL2.5-8B, and Qwen2.5-VL-7B reveals that visual neurons are significantly over-represented among hallucination-driving neurons (fold 1.10–1.35 $\times$, $p < 0.001$, Fisher’s exact test). Finally, activation steering on LLaVA-OneVision-7B demonstrates that scaling visual neurons by $\alpha = 0.5$ improves POPE from 90.2% to 92.0% (+1.8) and reduces CHAIR_i from 26.1% to 24.3% (−1.8) at inference time.

Our contributions are as follows:

- We propose PMBT, a non-parametric permutation test that replaces four hand-tuned thresholds with a single significance level, providing calibrated confidence scores for neuron modality classification.
- We validate the taxonomy across three VLM architectures via ablation, establishing a double dissociation between visual and text neuron function on six benchmarks.
- We identify hallucination-driving neurons via per-neuron ablation scoring and show that visual neurons are enriched among them in LLaVA-OneVision-7B, InternVL2.5-8B, and Qwen2.5-VL-7B (fold 1.10–1.35 $\times$, OR 1.15–1.93).

- We demonstrate that activation steering at $\alpha = 0.5$ reduces hallucination on LLaVA-OneVision-7B, providing a practical inference-time intervention.

2 Related Work

Vision-Language Models. Modern VLMs converge on a shared design: a pre-trained vision encoder extracts patch-level features that are projected into the token space of an autoregressive LLM backbone via a learnable connector. LLaVA [11] established this paradigm with a simple linear projection from CLIP ViT-L/14 [12] to Vicuna-7B, demonstrating that visual instruction tuning on a modest dataset yields strong multimodal capabilities. Subsequent work has scaled this recipe along three axes. LLaVA-OneVision [9] introduces dynamic-resolution encoding and cross-scenario transfer across images, multi-image, and video tasks. InternVL 2.5 [3] replaces CLIP with a proprietary InternViT encoder and scales to 78B parameters with improved training recipes. Qwen2.5-VL [1] adopts a native ViT encoder with dynamic resolution and achieves state-of-the-art results across 18 benchmarks. Alternative encoder choices include SigLIP [16], which replaces the contrastive softmax loss with a pairwise sigmoid, yielding improved scaling behaviour. Despite architectural diversity, different backbones, encoders, and projectors, all these models share the property that visual and textual information is processed jointly in the LLM’s feed-forward layers. This shared processing motivates the neuron-level analysis we pursue: if modality specialisation emerges in the FFN layers, it should be detectable across architectures regardless of the upstream encoder.

Hallucination in VLMs. VLMs frequently generate objects, attributes, or spatial relations not present in the input image, a phenomenon termed object hallucination. Li *et al.* [10] introduced POPE, a polling-based benchmark that probes hallucination via yes/no questions at three difficulty levels (random, popular, adversarial). Rohrbach *et al.* [13] proposed CHAIR, which measures the fraction of generated caption objects absent from the ground-truth scene graph. Together, POPE and CHAIR have become the standard evaluation protocol for VLM hallucination. Mitigation strategies operate at multiple levels. At the decoding level, Visual Contrastive Decoding (VCD) [8] amplifies the divergence between outputs conditioned on original versus distorted images, suppressing objects that the model generates regardless of visual input. OPERA [7] penalises attention patterns associated with over-trust in language priors and retrospectively reallocates attention to visual tokens. Training-level approaches include reinforcement learning from human feedback on hallucination-annotated data and retrieval-augmented generation. However, all these methods treat the model as a black box, intervening at the output or attention level without identifying which internal components are responsible. Our work complements this literature by identifying the neuron-level substrate of hallucination: we show that specific MH-Neurons are causally responsible and that their targeted suppression reduces hallucination, providing a mechanistic alternative to output-level correction.

Neuron-Level Interpretability. Understanding the function of individual neurons in deep networks has a long history. Network Dissection [2] pioneered concept-level neuron analysis in CNNs by measuring alignment between neuron activations and semantic segmentation masks. In LLMs, Dai *et al.* [4] identified knowledge neurons that store factual associations, showing that suppressing a small set of neurons erases specific facts without degrading general performance. Tang *et al.* [14] extended this approach to multilingual settings, discovering language-specific neurons whose activation or deactivation can steer the output language of LLMs, a direct analogue of our modality-specific steering. Most relevant to our work, Gao *et al.* [6] introduced H-Neurons, demonstrating that fewer than 0.1% of neurons in text-only LLMs can predict and causally trigger hallucination. However, their analysis is limited to unimodal LLMs and does not consider the multimodal setting where visual and textual information compete within shared layers.

For VLMs specifically, Xu *et al.* [15] pioneered the classification of LLM backbone neurons as visual, text, or multimodal based on activation patterns across modalities. Their taxonomy revealed that dedicated visual neurons exist even deep in the LLM backbone, far from the vision encoder. Fang *et al.* [5] proposed neuron attribution methods for multimodal LLMs, tracing model outputs to individual neurons via gradient-based attribution scores. However, Xu *et al.*'s classification relies on four hand-tuned thresholds (activation magnitudes and minimum token counts) that lack statistical justification, and neither work connects the modality taxonomy to hallucination. Our work extends Xu *et al.* in three ways: (i) we replace their fixed thresholds with a statistically principled permutation test (PMBT), (ii) we validate the taxonomy causally through ablation across three architectures, and (iii) we introduce MH-Neurons, the intersection of modality-specialised and hallucination-driving neurons, establishing a direct mechanistic link between modality processing and hallucination.

3 Methods

We present a pipeline for classifying the functional roles of individual neurons within the LLM backbone of VLMs and for identifying neurons that causally drive hallucination. Our approach extends the neuron taxonomy of Xu *et al.* [15] in three ways: (i) we replace the fixed-threshold classification with a non-parametric permutation test, (ii) we validate the taxonomy through ablation studies establishing a double dissociation, and (iii) we introduce a hallucination analysis that cross-references modality labels with neuron-level hallucination scores. An overview is shown in Fig. 1.

3.1 Preliminaries

Architecture. We focus on decoder-only VLMs sharing a common design: a vision encoder projects image patches into visual tokens, which are concatenated with text tokens and fed to an autoregressive LLM backbone. Following [15], we

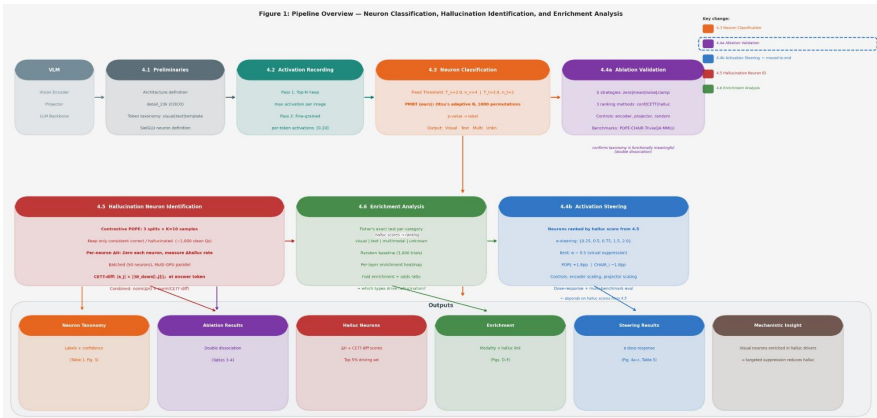


Fig. 1: Pipeline overview. Top row: Preliminaries (4.1), activation recording (4.2), neuron classification via FT and PMBT (4.3), and ablation validation confirming a double dissociation (4.4a). Bottom row: hallucination neuron identification via contrastive POPE and per-neuron ΔH scoring (4.5), enrichment analysis linking modality to hallucination drivers (4.6), and activation steering using hallucination scores to rank neurons for targeted suppression (4.4b). Steering is placed after enrichment because it depends on hallucination scores from 4.5.

define a *neuron* as one dimension of the activation vector produced by the gated activation in each FFN layer. In a SwiGLU FFN:

$$a_j = \text{SiLU}(W_{\text{gate}} \cdot x)_j, \quad j = 1, \dots, d_{\text{ffn}} \quad (1)$$

where W_{gate} is the gate projection matrix. For LLaVA-1.5 (Vicuna-7B), $d_{\text{ffn}} = 11,008$ neurons per layer across 32 layers (352,256 total). For Qwen2-based models, $d_{\text{ffn}} = 18,944$ across 28 layers (530,432 total).

Dataset. Following [15], we use the `detail_23k` subset of COCO train2017 (23,000 images). For each image, the VLM generates a description (100–300 tokens) via greedy decoding. The full prompt-plus-response sequence is replayed via teacher forcing to record neuron activations.

Token taxonomy. The input sequence contains three token types: (i) *template tokens* (BOS, role markers, prompt text), constant across images and excluded from analysis; (ii) *visual tokens* (projected image patches; 576 for LLaVA-1.5, variable for dynamic-resolution models); and (iii) *text tokens* (the generated description). Only visual and text tokens contribute to classification.

3.2 Neuron Activation Recording

Activation recording proceeds in two passes following [15].

Top- N heap. The first pass identifies, for each neuron, the $N = 50$ images that most strongly activate it. Each neuron’s per-image score is the maximum

activation across content token positions:

$$s_j^{(l)}(i) = \max_{t \in \text{content}} a_j^{(l)}(t) \quad (2)$$

A min-heap of size N per neuron maintains the top-scoring images. Simultaneously, each neuron’s global maximum activation is tracked for normalisation.

Activation pattern recording. The second pass re-processes only the top- N images and records fine-grained per-token activations. Each activation is divided by the neuron’s global maximum and rescaled to $[0, 10]$, with values clamped to this range.

3.3 Neuron Classification

Fixed-threshold classification [15]. For each neuron’s top- N samples, a per-sample modality label is assigned: visual-activated if $\geq n_v = 4$ visual tokens exceed $T_v = 2.0$, and text-activated if $\geq n_t = 2$ text tokens exceed $T_t = 3.0$. The neuron label is the argmax of proportions p_v, p_t, p_m, p_u over the N samples.

Permutation-based modality test (PMBT). For each neuron, all normalised per-token activations from the top- N samples are concatenated into vectors $\mathbf{v}$ (visual, length m) and $\mathbf{t}$ (text, length n). Otsu’s method finds the threshold θ minimising within-class variance on the activation histogram over $[0, 10]$. Let h_{vis} and h_{txt} denote the number of visual and text tokens exceeding θ . The test statistic is:

$$D = \frac{h_{\text{vis}}}{m} - \frac{h_{\text{txt}}}{n} \quad (3)$$

The null distribution is constructed by randomly permuting modality labels 1,000 times. The two-tailed p -value is the fraction of null permutations producing $|D|$ at least as extreme as observed. The neuron is classified as *visual* if $p < \alpha$ and $D > 0$; *text* if $p < \alpha$ and $D < 0$; *multimodal* if $p \geq \alpha$; or *unknown* if fewer than 5 tokens exceed θ . We set $\alpha = 0.05$.

3.4 Ablation Validation

Mechanism. During inference, forward hooks registered at each FFN’s gated activation function intercept neuron outputs. In ablation mode, targeted activations are set to zero.

Neuron ranking. Neurons are ranked by: (i) classification confidence ($1 - p$ -value), (ii) CETT (mean activation $\times \|W_{\text{down}}[:, j]\|_2$), or (iii) hallucination score from Sec. 3.5.

Ablation curve. We sweep top- $n = \{100, 500, 1K, 5K, \text{all}\}$, always in rank order.

Controls. Random baseline (count-matched), encoder ablation (α -scaled vision encoder MLPs), and projector ablation (α -scaled modality projector).

Benchmarks. POPE, CHAIR, VQAv2 (visual); TriviaQA, MMLU (text-only).

3.5 Hallucination Neuron Identification

Contrastive POPE filtering. For each POPE question across three splits, we sample $K = 10$ responses at temperature 0.7 and retain only consistently correct/wrong questions ($\sim 1,000$ clean questions).

Per-neuron ΔH . Each neuron is zeroed via a forward hook; the change in hallucination rate defines ΔH . Neurons are ablated in batches of 50, parallelised across GPUs.

Answer-token CETT-diff. $\text{CETT}_j = |a_j| \times \|W_{\text{down}}[:, j]\|_2$ at the answer token. $\text{CETT-diff}_j = \text{mean}(\text{CETT}_j \mid \text{halluc}) - \text{mean}(\text{CETT}_j \mid \text{correct})$.

Combined score. $\text{normalise}(\Delta H) \times \text{normalise}(\text{CETT-diff})$. Top $K\% = 5$ designated hallucination-driving.

3.6 Enrichment Analysis

For each modality category, we construct a 2×2 contingency table and apply Fisher’s exact test. Fold enrichment > 1 with $p < 0.05$ indicates over-representation among hallucination drivers. We calibrate with 1,000 random baseline trials and compute per-layer enrichment heatmaps.

Code availability. All code for neuron classification, ablation, hallucination scoring, and activation steering will be made publicly available upon acceptance.

3.7 Activation Steering

Beyond binary ablation, we evaluate activation steering by scaling targeted neurons’ activations by $\alpha \in \{0.25, 0.5, 0.75, 1.5, 2.0\}$, preserving activation patterns while modulating magnitude. Neurons are ranked by the combined hallucination score from Sec. 3.5, so the most hallucination-driving neurons are steered first. Encoder and projector controls test whether hallucination originates upstream of the LLM backbone.

4 Experiments and Results

4.1 Experimental Setup

Models. LLaVA-OneVision-7B (Qwen2, SigLIP, 530,432 neurons), InternVL2.5-8B (InternLM2, InternViT, 458,752 neurons), Qwen2.5-VL-7B (Qwen2, native ViT, 530,432 neurons). LLaVA-1.5-7b (352,256 neurons) serves as validation model. Full pipeline on LLaVA-OV; taxonomy and enrichment on all three.

Implementation. NVIDIA A100 80GB GPUs. Activation recording sharded by layer range. Permutation test vectorised on GPU ($\sim 100\times$ speedup). Ablation curves sweep 0.1–5% of neurons per category.

Table 1: Overall neuron type distribution (%) across models and classification methods.

Model	Method	Visual	Text	Multi.	Unk.
LLaVA-OV-7B	Fixed	48.9	7.0	31.7	12.4
	Perm.	45.2	34.1	5.8	14.9
InternVL2.5-8B	Fixed	46.7	8.0	24.8	20.5
	Perm.	36.3	52.6	11.1	0.0
Qwen2.5-VL-7B	Fixed	42.2	23.6	15.8	18.4
	Perm.	41.8	38.1	8.0	1.7

Table 2: Impact of classification method on neuron category proportions (Δ = Permutation – Fixed Threshold, in percentage points).

Model	Δ Visual	Δ Text	Δ Multi.	Δ Unk.
LLaVA-OV-7B	−3.7	+27.1	−25.9	+2.5
InternVL2.5-8B	−10.4	+44.6	−13.7	−20.5
Qwen2.5-VL-7B	−0.4	+14.5	−7.8	−16.7

4.2 Overall Neuron Distribution

Table 1 shows the overall neuron type distribution under both methods. Under the fixed threshold, multimodal neurons account for 15–32% across models. PMBT consistently deflates this category (to 5.8–11.1%), redistributing neurons primarily into the text category. InternVL2.5-8B shows the most dramatic redistribution: text neurons surge from 8.0% to 52.6% (6.6×). The unknown category is virtually eliminated under PMBT for InternVL (20.5% → 0.0%) and Qwen (18.4% → 1.7%).

Table 2 quantifies the shift when moving from FT to PMBT. The most consistent effect is the reduction in multimodal neurons (−25.9, −13.7, and −7.8 percentage points for LLaVA-OV, InternVL, and Qwen respectively). InternVL2.5-8B shows the most dramatic redistribution, with text neurons surging by +44.6. This suggests that many neurons classified as multimodal by FT are, under rigorous testing, better characterised as modality-specific with weak cross-modal responses that do not survive permutation-based significance testing.

Figure 2 illustrates this reclassification qualitatively for six neurons in LLaVA-1.5-7b. Panels (a)–(c) show agreement: both methods assign visual (a), text (b), and visual (c) respectively. The disagreements are more instructive. Panel (d) is a case where FT arguably wins: this neuron activates on both confectionery images and descriptive tokens, and FT’s multimodal label ($p_m = 0.82$) appears reasonable given the mixed evidence. PMBT reclassifies it as visual, but with a near-miss: $D = +0.012$, $p = 0.031$, barely passing $\alpha = 0.05$. Panels (e,f) show the reverse: both neurons activate primarily on descriptive text tokens (zebra descriptions, urban scene narration), yet FT labels them multimodal with high



Fig. 2: Qualitative reclassification examples for six neurons in LLaVA-1.5-7b (selected to match Xu *et al.* [15], their Fig. 3). Each panel shows the top-activating images, activation patterns over visual (green) and text tokens, and the generated description with highlighted tokens. Top row: (a) visual neuron, (b) text neuron, (c) multimodal→visual under PMBT. Bottom row: (d) multimodal→visual, (e,f) multimodal→text under PMBT.

confidence ($p_m = 0.98$). PMBT correctly identifies them as text ($D = -0.12$, $p = 0$). The score on disagreements is thus 1–1, but the failure modes differ qualitatively. PMBT’s error on (d) is a borderline call: D is near zero and p barely passes the threshold, signalling low confidence that is directly readable from the test statistics. FT’s errors on (e,f) are hidden behind high probabilities ($p_m = 0.98$), giving no indication that the classification is wrong. This illustrates a key qualitative advantage of PMBT: its uncertainty is transparent in D and p , whereas FT’s errors are masked by overconfident probability estimates.

4.3 Classification Confidence and Statistical Validity

The classification probability distributions and permutation test statistics provide insight into the confidence of neuron assignments under each method.

Fixed-threshold probability distributions. Under FT, the visual probability distribution shows a strong rightward skew, with a large concentration of neurons near probability 1.0, indicating that many neurons are classified as visual with high confidence. The text and multimodal distributions are more uniform or left-skewed, suggesting weaker and more ambiguous assignments for these categories. This asymmetry is consistent across all four models.

Permutation test statistics. The p -value distributions show a characteristic bimodal pattern: a large spike near $p = 0$ (neurons with strong modality preference) and a secondary accumulation near $p = 1$ (neurons with no detectable preference). The observed rate difference histograms are sharply concentrated near

zero, indicating that most neurons have only small absolute differences in activation between modalities, with the classification hinging on whether this small difference is statistically significant. The confidence-by-label histograms reveal that visual and text neurons tend to be classified with high confidence (clustered near 1.0), while multimodal neurons exhibit a broader confidence distribution, consistent with inherently noisier cross-modal responses. Per-neuron confidence histograms and rate-difference distributions for all models are in Supp. Sec. D.

4.4 Cross-Model Comparative Analysis

Figure 3 presents the per-layer neuron type distributions under both methods. Under FT (top row), all three models assign a substantial green (multimodal) band, but the magnitude varies: LLaVA-OV (centre) shows the most dramatic multimodal dominance in layers 0–12; InternVL (left) distributes more evenly; Qwen (right) has a large unknown band in early layers. Under PMBT (bottom row), InternVL is text-dominated (blue >50%), LLaVA-OV shows a clean text-to-visual transition at layer 7–8, and Qwen displays a balanced split. All three models share sharper layer boundaries under PMBT, where the multimodal band previously masked the underlying structure.

Architectural influence on neuron distribution. Under FT, InternVL2.5-8B shows a relatively even split between visual, text, and multimodal neurons, with a pronounced unknown band in layers 0–2 that vanishes under PMBT. LLaVA-OV shows multimodal dominance in layers 0–12, with visual neurons growing only in deeper layers. Qwen has a large unknown band in layers 0–3 under FT that gives way to visual dominance from layer 5. Under PMBT, InternVL becomes text-dominated (blue >50% in most layers), LLaVA-OV shows a clean text-to-visual transition (blue dominates layers 0–7, red dominates 8–27), and Qwen displays the most balanced visual–text split.

Layer-wise specialisation patterns. Under PMBT, all three models share a common organisational principle: early layers are more heterogeneous while deeper layers show increasing modality specialisation. In LLaVA-OV, the first seven layers are heavily text-dominated, transitioning sharply to visual-dominated around layer 7–8. InternVL shows text dominance throughout but with visual neurons increasing from ~25% in early layers to ~35% in late layers. Qwen displays the most gradual shift, with visual growing from ~30% at layer 0 to ~60% at layer 27.

Sensitivity to classification method. The most dramatic transformation occurs in LLaVA-OV, where the extensive multimodal band under FT is almost entirely eliminated under PMBT. InternVL shows the largest shift in the visual–text balance: text neurons surge from 8.0% to 52.6% (a $6.6\times$ increase). Qwen is the most stable in its visual proportion between methods, but its unknown category (prominent in FT layers 0–3) is virtually eliminated under PMBT.

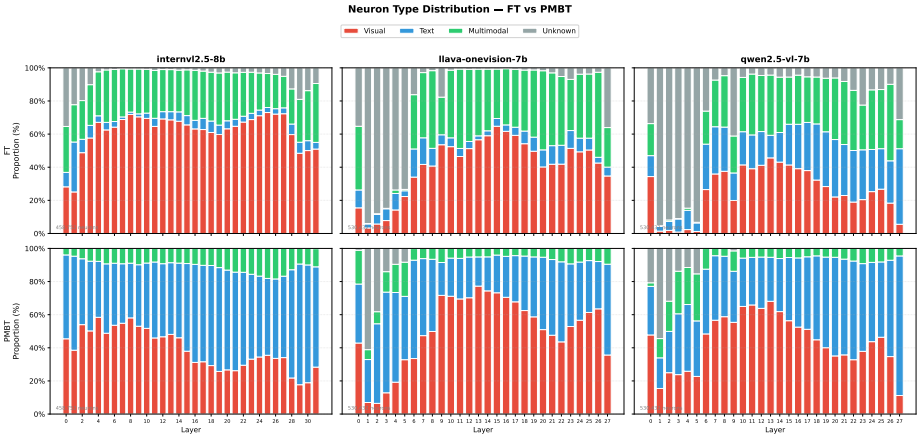


Fig. 3: Per-layer neuron type distribution for InternVL2.5-8B (left), LLaVA-OV-7B (centre), and Qwen2.5-VL-7B (right) under fixed threshold (top, FT) and permutation test (bottom, PMBT). The multimodal band (green) collapses from top to bottom row across all models.

4.5 Modality Taxonomy Ablation

Table 3 shows full-category ablation results for all three models. Across all architectures, visual neuron ablation produces catastrophic failure: POPE drops to chance (50.0%) in all three models. Text ablation selectively degrades language benchmarks while partially preserving POPE. Notably, InternVL2.5-8B shows catastrophic multimodal ablation (POPE: 50.0%), suggesting multimodal neurons serve as integration hubs in that architecture, while in LLaVA-OV and Qwen multimodal ablation causes only modest degradation. The double dissociation is preserved across all three models: visual ablation preferentially degrades vision benchmarks, text ablation preferentially degrades language benchmarks. The random baseline degrades less selectively than targeted ablation, confirming the taxonomy captures functionally coherent populations. InternVL and Qwen ablation results with F1 and CHAIR_s are in Supp. Sec. E.

Scaled ablation: sensitivity to ablation fraction. To understand the dose-response relationship, we ablate only the top 0.1%, 0.5%, 1%, and 5% of neurons within each category for LLaVA-OV (ranked by classification confidence).

At 0.1–1.0% ablation, all categories show only minor performance changes (<1 on POPE/F1), indicating substantial within-category redundancy (full table in Supp. Sec. F). A sharp non-linear transition occurs between 1% and 5%: visual TQA degrades from -0.25 at 1% to -8.6 at 5%; text MMLU from -0.05 at 1% to -8.9 at 5%. This non-linearity suggests a critical threshold beyond which neuron redundancy is exhausted. Visual ablation preferentially impacts POPE and CHAIR_i (vision-dependent), while text ablation most severely degrades TQA and MMLU (language-dependent), providing cross-validated evidence that the taxonomy captures genuine functional specialisation.

Table 3: Full-category ablation results across all three models. Bold indicates catastrophic degradation (>10 absolute drop).

Model Condition		% Abl.	POPE	CHAIR _i	TQA	MMLU
LLaVA-OV	<i>Baseline</i>	—	<i>90.2</i>	<i>26.1</i>	<i>48.2</i>	<i>64.1</i>
	Abl. Visual	51.1	50.0	19.5	2.0	19.1
	Abl. Text	36.2	80.2	25.0	9.3	38.1
	Abl. Multi.	8.0	89.6	24.5	40.1	60.3
	Abl. Unk.	4.7	88.5	25.8	44.0	60.6
	Random	36.2	87.1	16.7	9.1	34.2
InternVL	<i>Baseline</i>	—	<i>93.8</i>	<i>20.8</i>	<i>71.3</i>	<i>72.4</i>
	Abl. Visual	37.8	50.0	0.0	0.0	3.8
	Abl. Text	51.5	52.6	25.0	0.7	28.7
	Abl. Multi.	10.7	50.0	0.0	0.5	11.5
	Abl. Unk.	0.0	93.7	20.8	70.3	72.9
	Random	51.5	50.0	0.0	0.0	0.0
Qwen	<i>Baseline</i>	—	<i>84.4</i>	<i>26.9</i>	<i>63.6</i>	<i>68.9</i>
	Abl. Visual	43.8	50.0	83.2	0.0	0.0
	Abl. Text	41.8	63.1	0.0	0.5	0.1
	Abl. Multi.	9.0	83.5	25.8	49.1	61.9
	Abl. Unk.	5.3	82.1	27.3	54.3	57.6
	Random	41.8	87.2	26.3	7.6	34.6

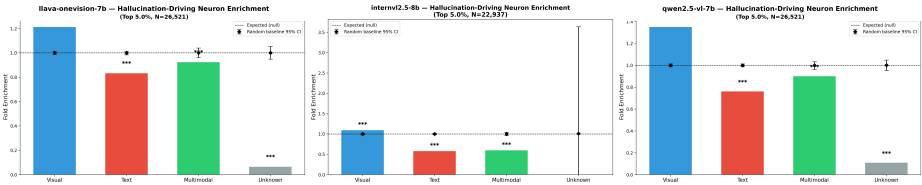


Fig. 4: Hallucination-driving neuron enrichment (top 5% by ΔH) for LLaVA-OneVision-7B (left), InternVL2.5-8B (centre), and Qwen2.5-VL-7B (right). Bars show fold enrichment per modality category; dashed line = expected null ($1.0\times$); diamonds = random baseline 95% CI. Visual neurons (blue) are significantly enriched in all three models ($1.10\text{--}1.35\times$, $p < 0.001$), while text (red) and unknown (grey) are depleted. *** = $p < 0.001$, Fisher’s exact test.

4.6 Enrichment Analysis Results

Visual MH-Neurons are significantly enriched among hallucination-driving neurons in all three models (Fig. 4). In LLaVA-OneVision-7B (left panel), visual neurons show fold enrichment of $1.21\times$ ($\text{OR}=1.60$), exceeding both the expected null and the random baseline 95% CI. Text neurons are depleted (fold = $0.83\times$) and unknown neurons are near-absent (fold = $0.07\times$). InternVL2.5-8B (centre) shows a similar pattern with visual enrichment of $1.10\times$ ($\text{OR}=1.15$), but with stronger text and multimodal depletion (fold = $0.58\times$ and $0.60\times$ respectively), consistent with the catastrophic multimodal ablation observed in Tab. 3. Qwen2.5-VL-7B (right) exhibits the strongest visual enrichment at $1.35\times$ ($\text{OR}=1.93$). Across all three models, unknown neurons are near-absent from the hallucination-driving set (fold ≤ 0.11 , $\text{OR} \leq 0.10$), confirming that neurons PMBT cannot confidently classify are also not driving hallucination. All results are significant at

Table 4: Steering results at $\alpha = 0.5$ (Δ vs. baseline) for LLaVA-OneVision-7B. Baseline: POPE=0.902, CHAIR_i=0.261, TQA=0.482, MMLU=0.641.

Condition	Δ POPE	Δ CHAIR _i	Δ TQA	Δ MMLU
Visual	+0.018	-0.018	-0.125	-0.058
Text	-0.007	-0.025	-0.093	-0.012
Multimodal	-0.001	-0.013	-0.005	-0.007
Random	-0.009	-0.043	-0.096	-0.014

$p < 0.001$ (Fisher’s exact test). Per-layer heatmaps and odds ratio forest plots are in Supp. Sec. G.

Three-method comparison (LLaVA-OV). We additionally compare enrichment across three scoring methods: ΔH (ablation), CETT-diff, and the combined score. Visual neurons are enriched by ΔH (fold = $1.21\times$) and mildly by CETT-diff ($1.07\times$) and combined ($1.06\times$). Interestingly, CETT-diff shows text neurons slightly enriched ($1.09\times$) while ΔH shows them depleted ($0.83\times$). This divergence suggests that text neurons contribute substantially to the hidden state during hallucination (high CETT) but their removal does not increase hallucination rate (low ΔH), consistent with a compensatory rather than causal role. The agreement on visual enrichment across all three methods strengthens the causal interpretation.

4.7 Activation Steering Results

We report steering results for LLaVA-OV (Tab. 4, Fig. 5). Visual MH-Neuron suppression at $\alpha = 0.5$ improves POPE from 90.2% to 92.0% (+1.8) and reduces CHAIR_i from 26.1% to 24.3% (-1.8), peaking at $\alpha = 0.5$ in dose-response curves. Amplification ($\alpha > 1$) worsens both, dropping POPE to $\sim 55\%$ at $\alpha = 3.0$, a bidirectional causal link. Text suppression at $\alpha = 0$ collapses POPE to $\sim 80\%$ and CHAIR_i to $\sim 0\%$ (the model generates almost nothing). Multimodal suppression has negligible impact ($\Delta < 1.3$). The random baseline degrades CHAIR_i more (-4.3) without POPE improvement, confirming taxonomy-guided steering outperforms undirected disruption.

Encoder and projector controls. Encoder scaling produces catastrophic failure: POPE collapses to chance ($\sim 50\%$) and CHAIR_i spikes to $\sim 75\%$ regardless of α , confirming the encoder is indispensable. Projector scaling is less severe but still degrades performance at extreme α values (POPE drops to $\sim 85\%$ at $\alpha = 2.5$). In contrast, LLM-level visual neuron suppression at $\alpha = 0.5$ improves POPE and reduces CHAIR_i simultaneously. This localises the hallucination mechanism to the LLM backbone’s processing of visual features, not to the visual features themselves or the cross-modal projection.

Ablation curve at $\alpha = 0.5$. Sweeping the number of top- n neurons suppressed at $\alpha = 0.5$ reveals that visual neurons produce the steepest initial POPE degradation per neuron (from 90.2% to 89.3% when all visual neurons are sup-

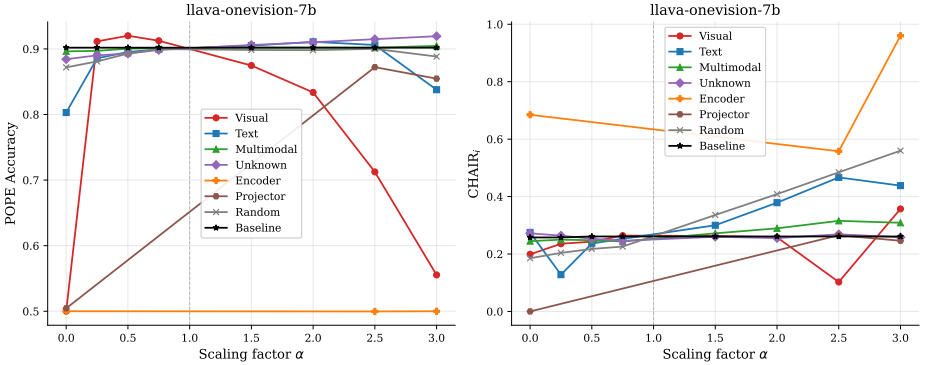


Fig. 5: Dose-response curves for LLaVA-OV-7B. Left: POPE accuracy vs. α . Right: CHAIR_i vs. α . Visual suppression ($\alpha < 1$) improves both; amplification worsens both. Encoder scaling is catastrophic.

pressed), consistent with their enrichment among hallucination drivers. The random baseline drops more steeply at high n , while multimodal suppression has almost no effect throughout. Unknown neurons show a sharp drop only when all are suppressed, suggesting a concentrated rather than distributed effect. Multi-benchmark bar chart and ablation curves are in Supp. Sec. H. Steering results for InternVL2.5-8B and Qwen2.5-VL-7B are provided in Supp. Sec. J.

5 Conclusion

We presented a pipeline for classifying and validating neuron roles in VLM backbones and connecting modality specialisation to hallucination. Our permutation-based modality test (PMBT) replaces four hand-tuned hyperparameters with a single significance level, reclassifying 15–32% of neurons from multimodal to modality-specific categories across four architectures. Ablation confirms a double dissociation: visual neuron ablation degrades vision benchmarks while text ablation degrades language benchmarks. Enrichment analysis on LLaVA-OneVision-7B, InternVL2.5-8B, and Qwen2.5-VL-7B reveals visual MH-Neurons are over-represented among hallucination drivers (fold 1.10–1.35 $\times$, $p < 0.001$), and steering at $\alpha = 0.5$ on LLaVA-OneVision-7B improves POPE from 90.2% to 92.0% and reduces CHAIR_i from 26.1% to 24.3%, localising hallucination to the LLM backbone. Cross-architecture steering results are in Supp. Sec. J. Future work includes combining MH-Neuron identification with decoding strategies such as VCD [8] and OPERA [7], and taxonomy-informed architectural modifications.

References

1. Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J.: Qwen2.5-VL technical report. arXiv preprint arXiv:2502.13923 (2025) 1, 3
2. Bau, D., Zhou, B., Khosla, A., Oliva, A., Torralba, A.: Network dissection: Quantifying interpretability of deep visual representations. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 6541–6549 (2017) 4
3. Chen, Z., Wang, W., Cao, Y., Liu, Y., Gao, Z., Cui, E., Zhu, J., Ye, S., Tian, H., Liu, Z.: Expanding performance boundaries of open-source multimodal models with model, data, and test-time scaling. arXiv preprint arXiv:2412.05271 (2024) 1, 3
4. Dai, D., Dong, L., Hao, Y., Sui, Z., Chang, B., Wei, F.: Knowledge neurons in pretrained transformers. In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 8493–8502 (2022). <https://doi.org/10.18653/v1/2022.acl-long.581> 4
5. Fang, J., Bi, Z., Wang, R., Jiang, H., Gao, Y., Wang, K., Zhang, A., Shi, J., Wang, X., Chua, T.S.: Towards neuron attributions in multi-modal large language models. In: Advances in Neural Information Processing Systems (NeurIPS). vol. 37, pp. 122867–122890 (2024) 4
6. Gao, C., Chen, H., Xiao, C., Chen, Z., Liu, Z., Sun, M.: H-Neurons: On the existence, impact, and origin of hallucination-associated neurons in LLMs. arXiv preprint arXiv:2512.01797 (2025) 4
7. Huang, Q., Dong, X., Zhang, P., Wang, B., He, C., Wang, J., Lin, D., Zhang, W., Yu, N.: OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13418–13427 (2024) 3, 14
8. Leng, S., Zhang, H., Chen, G., Li, X., Lu, S., Miao, C., Bing, L.: Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13763–13773 (2024) 3, 14
9. Li, B., Zhang, Y., Guo, D., Zhang, R., Li, F., Zhang, H., Zhang, K., Li, Y., Liu, Z., Li, C.: LLaVA-OneVision: Easy visual task transfer. In: Advances in Neural Information Processing Systems (NeurIPS) (2024) 3
10. Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W.X., Wen, J.R.: Evaluating object hallucination in large vision-language models. In: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 292–305 (2023) 3
11. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 26296–26306 (2024) 1, 3
12. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Proceedings of the 38th International Conference on Machine Learning (ICML). pp. 8748–8763 (2021) 3
13. Rohrbach, A., Hendricks, L.A., Burns, K., Darrell, T., Saenko, K.: Object hallucination in image captioning. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 4035–4045 (2018) 3

14. Tang, T., Luo, W., Huang, H., Zhang, D., Wang, X., Zhao, X., Wei, F., Wen, J.R.: Language-specific neurons: The key to multilingual capabilities in large language models. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL). pp. 5701–5715 (2024). <https://doi.org/10.18653/v1/2024.acl-long.309>
15. Xu, J., Lan, C., Lu, Y.: Deciphering functions of neurons in vision-language models. In: Proceedings of the 33rd ACM International Conference on Multimedia (MM '25). pp. 1–9 (2025). <https://doi.org/10.1145/3746027.3755003>
16. Zhai, X., Mustafa, B., Kolesnikov, A., Beyer, L.: Sigmoid loss for language image pre-training. In: Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). pp. 11975–11986 (2023)